

# Yuhang Li

albertl06011@gmail.com · +86 180 4013 1985 · github.com/yhli-ml · Google Scholar

Research in **trustworthy machine learning and foundation models**, with first-author papers at **ICLR 2025** and **ICCV 2025**; experience in weak supervision and LLM inference systems.

## Education

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**The University of Tokyo** Oct. 2026 – Present

*Ph.D. Student, Complexity Science and Engineering*

Graduate School of Frontier Sciences; advisor: Prof. Masashi Sugiyama.

**Southeast University** Sep. 2023 – Jun. 2026

*M.Eng. in Computer Science and Technology*

Advisor: Prof. Yuheng Jia, PAttern Learning and Mining (PALM) Lab.

**Dalian University of Technology** Sep. 2019 – Jun. 2023

*B.Eng. in Computer Science and Technology*

## Research Experience

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**Huawei, 2012 Laboratories** Jun. 2026 – Oct. 2026

*Research Intern, Theory Research Department*

- Developed **request-arrival-triggered prefix KV-cache prefetching** in vLLM + LMCache for long-context inference across GPU memory, host memory, and SSD storage. Moved SSD-to-host transfers into the request’s queueing window to overlap I/O with ongoing computation.
- The complete implementation delivered **31.57% higher request throughput** and **24.49% lower mean time to first token (TTFT)** than native prefetch in a 3,379-request replay using a 12-layer validation model and one output token per request. The related **patent disclosure passed internal review**.
- Implemented and validated asynchronous prefix prefetching in **vLLM-Ascend + Unified Cache Management (UCM)**, covering cache residency protection, shared I/O and multi-rank coordination, and cancellation/fallback. Validated both MLA and standard GQA paths while reusing native scheduling and device KV loading.

**RIKEN Center for Advanced Intelligence Project** Dec. 2025 – May 2026

*Student Trainee, Imperfect Information Learning Team* Tokyo, Japan

- Conducted research on weakly supervised learning with foundation models, focusing on long-tailed distributions and noisy labels, advised by **Prof. Masashi Sugiyama**.

## Selected Publications

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1. **Complementary Label Learning with Positive Label Guessing and Negative Label Enhancement.**

**Yuhang Li**, Zhuying Li, Yuheng Jia. **ICLR 2025**. [paper] [code]

Formulated complementary-label learning as an inverse problem, combining positive-label guessing with negative-label enhancement to recover supervision. On STL-10 with one complementary label per example, achieved **55.25% accuracy vs. POCL’s 34.96% (+20.29 percentage points)**.

2. **Boosting Class Representation via Semantically Related Instances for Robust Long-Tailed Learning with Noisy Labels.**

**Yuhang Li**, Zhuying Li, Yuheng Jia. **ICCV 2025**. [paper] [code]

Designed semantic soft-label transfer and a shot-specific multi-expert ensemble. On CIFAR-10 with 50% label noise and imbalance ratio 200, achieved **69.80% top-1 accuracy vs. SFA’s 64.81% (+4.99 percentage points)**.

3. **Samples Are Not Equal: A Sample Selection Approach for Deep Clustering.**

Zhengxing Jiao, Yaxin Hou, Jun Ma, **Yuhang Li**, Ding Ding, Yuheng Jia, Hui Liu, Junhui Hou. **ICLR 2026**. [paper]

## Manuscript Under Review

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**Robust Long-Tailed Partial Label Learning by Head Classifier and Tail Classifier Cooperation.**

Jia, Y., **Li, Y.**, Peng, X., Wang, R., Zhang, M.

**Under review**; first student author (second in the author list). [manuscript]

Designed head/tail classifier cooperation to address class imbalance and label ambiguity.

## Selected Honors

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- **JSPS DC1 Research Fellowship**, FY2027: provisionally selected; appointment pending.
- **BOOST NAIS, The University of Tokyo (JST-supported)**, 2026. Provisionally selected; formal appointment pending.
- **National Scholarship for Graduates**, 2025.
- **SI-TECH “Future Star” Education Fund, Second Prize Scholarship**, 2025.

## Technical Skills & Languages

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**Research:** Python, PyTorch; weak supervision, robust and long-tailed learning.

**Inference systems:** vLLM, LMCache, vLLM-Ascend, UCM; KV-cache prefetching.

**Tools:** Linux, Git, Slurm.     **Languages:** English (IELTS 7.5), Mandarin (native).

## Academic Service

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Reviewer: NeurIPS 2026; ICLR 2026; IEEE MLMC 2025/2026; TMLR; Pattern Recognition; IJMLC.